

Lexicographical order :-

compareTo

Dictionary

bdz < c

s1.compareTo(s2);

String s1 = "bdz";

String s2 = "c";

s1 < s2 → -ve number

s1 == s2 → 0

s1 > s2 → +ve number

String s = "Hello";

H	e	l	l	o
0	1	2	3	4

s[0], s[1] X

s.charAt(1); ✓

Check Palindrome :-

↓
racecar ←

(a b i d e f)

psvm {

String s = "racecar";

boolean isPalindrome = true;

int left = 0, right = s.length() - 1;

while (left <= right) {

if (s.charAt(left) != s.charAt(right))

isPalindrome = false;

left++, right--;

}

Substring 1-

(contiguous part of string)

String $s = \text{"hello"}$
0 1 2 3 4

Sout (s.substring(1, 4)) ; ell

↓ inclusive → exclusive

hello
h e l l o
he el ell
hel ell ello
hell ello
hello

```
for (int i = 0; i < n; i++) {  
    for (int j = i; j < n; j++) {  
        Sout (s.substring(i, j+1));  
    }  
}
```


Lengthwise Substring :-

"abcde"

a | b | c | d | e |

Length 1 $\rightarrow$

a

b

c

d

e

2
 $\downarrow$

ab

bc

cd

de

3
 $\downarrow$
abc

bcd

cde

4
 $\downarrow$
abcd

bcd

5
 $\downarrow$
abcde

$\rightarrow$ len = 3

$$n - \text{len} = 5 - 3 = 2$$

$$5 - 1 + 1$$

$$5 - 2 + 1$$

$$5 - 3 + 1$$

xxx

very important

for each substring we

have $n - \text{len} + 1$ substrings.

Frequency Counting

$s = \text{leetcode}$

4. $\rightarrow \text{toCharArray}();$

$$y = [e^1, e^2, e^3, \dots, e^n]$$
$$l \rightarrow 1 \quad (\text{hell}^\circ)$$
$$e \rightarrow 3$$

九一

$$C \rightarrow I$$
$$0 \rightarrow 1$$
$$d \rightarrow 1$$

abb'dz -

2	3	2	5			...	25
0	1	2	3	4	5	...	25
a	b	c	d	e	f		

```
int[] freq = new int[26];
```

1	2				
1	1			1	1
0	4		d		2
(a)	b				

Convert char into
index $\rightarrow$
freq[chr - 'a']
 $\frac{r^h - 'a'}{b - 'a'}$
 $\downarrow$ 0 21

Book Allocation :-

25	46	28	49	24
----	----	----	----	----

K=4

49 →

1 → 25
2 → 46
3 → 28
4 → 49
5 → 24

Go → 1 → 25
2 → 46
⋮
⋮
⋮

71

$$1 + 25 + 46 = 71$$

2 → 28

3 → 49

4 → 24

48

is

$(25, 46, 28, 49, 24)$ ($k=4$)
~~25~~ ~~46~~ ~~28~~ ~~49~~ ~~24~~
 (Sum of all)

low = max(arr);

high = sum(arr);

(1)

Students

page student

~~1~~

~~2~~

~~3~~

4

~~25 + 46~~
~~71~~ 1

~~71 + 28~~ 2

28 + 49 = X

49 + 24 = X

25 46 28 49 24
~~25~~ ~~46~~ ~~28~~ ~~49~~ 24
~~1~~ ~~1~~ ~~1~~ ~~1~~ 1

(27)

Student → 73 → 24

71 72 73 74... (76)

